

SEAS 8520 - Deep Learning and Natural Language Processing

Homework 7 — Solution Key

Due Date: July 10, 2026 (11:59pm EST)

Questions (0.5 pts each)

Problem 1

Bounding boxes are given in (x, y, w, h) format, where (x, y) is the top-left corner and w, h are the width and height. A ground-truth box is $(1, 1, 4, 3)$ and a detector predicts $(3, 2, 4, 4)$. What is the Intersection over Union (IoU) of the prediction with the ground truth?

- (A) $1/4$
- (B) $1/3$
- (C) $1/6$
- (D) $1/7$

Step 1 (convert to corners): the ground-truth box spans $x \in [1, 5]$, $y \in [1, 4]$; the predicted box spans $x \in [3, 7]$, $y \in [2, 6]$.

Step 2 (box areas): ground truth $4 \times 3 = 12$; prediction $4 \times 4 = 16$.

Step 3 (intersection): overlap in x is $[3, 5]$ (width 2) and in y is $[2, 4]$ (height 2), so the intersection area is $2 \times 2 = 4$.

Step 4 (union): $12 + 16 - 4 = 24$.

Step 5 (IoU): $\frac{4}{24} = \frac{1}{6}$.

Problem 2

Using the fixed vocabulary $V = [\text{the, cat, sat, on, mat, dog, chased}]$, encode the document

“The cat sat on the mat. The dog chased the cat.”

as a *count* (term-frequency) bag-of-words vector after lowercasing the text and stripping punctuation. Which vector is correct?

- (A) [4, 2, 1, 1, 1, 1, 1]
- (B) [1, 1, 1, 1, 1, 1, 1]
- (C) [2, 2, 1, 1, 1, 1, 1]
- (D) [4, 1, 1, 1, 1, 1, 1]

Step 1 (standardize): lowercasing and stripping punctuation gives “the cat sat on the mat the dog chased the cat”.

Step 2 (count over the fixed vocabulary): the occurs 4 times, cat twice, and sat, on, mat, dog, chased once each.

Step 3 (assemble): in the given order the vector is [4, 2, 1, 1, 1, 1, 1]. Option (B) is binary presence, (C) forgets to lowercase the sentence-initial “The”, and (D) miscounts cat.

Problem 3

A corpus contains 10,000 reviews. In review d , the term *plot* occurs 5 times and appears in 100 reviews overall, while the term *film* occurs 8 times and appears in 1,000 reviews overall. Using term frequency (raw count) and base-10 inverse document frequency, what are the tf-idf weights of the two terms in d , and which term is more characteristic of d ?

- (A) *plot*: 8; *film*: 10 (*film* more characteristic)
- (B) *plot*: 10; *film*: 8 (*plot* more characteristic)
- (C) *plot*: 10; *film*: 16 (*film* more characteristic)
- (D) *plot*: 5; *film*: 8 (*film* more characteristic)

Step 1 (plot idf): $\log_{10} \frac{10000}{100} = \log_{10}(100) = 2$.

Step 2 (plot tf-idf): $5 \times 2 = 10$.

Step 3 (film idf): $\log_{10} \frac{10000}{1000} = \log_{10}(10) = 1$.

Step 4 (film tf-idf): $8 \times 1 = 8$.

Step 5 (compare): $10 > 8$, so although film occurs more often, plot is rarer across the corpus and earns the higher tf-idf weight, making it more characteristic of d .

Problem 4

A multinomial Naive Bayes spam filter is trained on the four messages below (word tokens listed with repetition):

Message	Word tokens	Label
1	win, win, cash, prize, win	spam
2	win, cash, prize	spam
3	cash, cash, cash, win	not spam
4	cash, win, prize, prize	not spam

Estimate the class priors and word probabilities by maximum likelihood (relative frequencies, no smoothing), then classify the new message $d = \text{"win cash cash"}$. What label does the model assign, and what is that label's unnormalized posterior score?

- (A) spam, with score $1/64$
- (B) not spam, with score $1/64$
- (C) spam, with score $1/32$
- (D) not spam, with score $1/32$

Step 1 (priors): 2 of 4 messages are spam, so $P(\text{spam}) = P(\text{not spam}) = \frac{1}{2}$.

Step 2 (spam word counts): messages 1–2 give win= 4, cash= 2, prize= 2, a total of 8 tokens, so $P(\text{win} \mid \text{spam}) = \frac{4}{8} = \frac{1}{2}$, $P(\text{cash} \mid \text{spam}) = \frac{2}{8} = \frac{1}{4}$.

Step 3 (not-spam word counts): messages 3–4 give cash= 4, win= 2, prize= 2, a total of 8 tokens, so $P(\text{win} \mid \text{not spam}) = \frac{2}{8} = \frac{1}{4}$, $P(\text{cash} \mid \text{not spam}) = \frac{4}{8} = \frac{1}{2}$.

Step 4 (score $d = \text{"win cash cash"}$): spam = $\frac{1}{2} \cdot \frac{1}{2} \cdot \left(\frac{1}{4}\right)^2 = \frac{1}{64}$; not spam = $\frac{1}{2} \cdot \frac{1}{4} \cdot \left(\frac{1}{2}\right)^2 = \frac{1}{32}$.

*Step 5 (decide): $\frac{1}{32} > \frac{1}{64}$, so the model labels d **not spam** with score $\frac{1}{32}$.*

Problem 5

Three 6-dimensional word embeddings are

$$\mathbf{a} = [0.2, 0.5, 0.1, 0.4, 0.0, 0.3],$$

$$\mathbf{b} = [0.1, 0.2, 0.4, 0.0, 0.5, 0.2],$$

$$\mathbf{c} = [0.4, 0.3, 0.2, 0.5, 0.1, 0.6].$$

A phrase vector is formed by summing $\mathbf{a}$ and $\mathbf{b}$. What is the cosine similarity between $(\mathbf{a} + \mathbf{b})$ and $\mathbf{c}$ (to two decimal places)?

- (A) 0.84
- (B) 0.79
- (C) 0.91
- (D) 0.67

Step 1 (sum): $\mathbf{a} + \mathbf{b} = [0.3, 0.7, 0.5, 0.4, 0.5, 0.5]$.

Step 2 (dot product with $\mathbf{c}$): $(0.3)(0.4) + (0.7)(0.3) + (0.5)(0.2) + (0.4)(0.5) + (0.5)(0.1) + (0.5)(0.6) = 0.12 + 0.21 + 0.10 + 0.20 + 0.05 + 0.30 = 0.98$.

Step 3 (norm of $\mathbf{a} + \mathbf{b}$): $\sqrt{0.09 + 0.49 + 0.25 + 0.16 + 0.25 + 0.25} = \sqrt{1.49} \approx 1.221$.

Step 4 (norm of $\mathbf{c}$): $\sqrt{0.16 + 0.09 + 0.04 + 0.25 + 0.01 + 0.36} = \sqrt{0.91} \approx 0.954$.

Step 5 (cosine): $\frac{0.98}{1.221 \times 0.954} = \frac{0.98}{1.165} \approx 0.84$.

Problem 6

Which one of the following statements about modern object detectors is correct?

- (A) Anchor boxes are predicted directly from raw pixels with no predefined shapes or aspect ratios.
- (B) Faster R-CNN is a single-stage detector that bypasses region proposals to run faster.
- (C) A region proposal network shares convolutional features with the detector, making Faster R-CNN trainable end to end.
- (D) Non-maximum suppression merges overlapping boxes by averaging their coordinates and confidence scores.

(C) is correct: the region proposal network reuses the backbone's features so Faster R-CNN trains end to end. Anchors are preset reference shapes (A), Faster R-CNN is two-stage (B), and non-maximum suppression discards overlapping lower-confidence boxes rather than averaging them (D).

Problem 7

Which statement about byte-pair encoding (BPE) tokenization is correct?

- (A) BPE begins with characters and repeatedly merges the most frequent adjacent pair, so unseen words back off to known subwords.
- (B) BPE assigns every distinct word a single integer id and maps anything unseen to one fixed out-of-vocabulary token.
- (C) BPE begins with a full-word vocabulary and splits rare words until only single characters remain.
- (D) BPE merges adjacent units only across word boundaries, so a space is required between every subword.

BPE builds subwords bottom-up by merging the most frequent adjacent pair, so any unseen word decomposes into known pieces instead of a single OOV token; (B) is word-level encoding, (C) reverses the direction, and merges stay within words.

Problem 8

The distributional hypothesis is the idea that underlies learned word embeddings. Which statement states it correctly?

- (A) A word's meaning is fixed by its dictionary definition, independent of how it is used in text.
- (B) Two words count as similar only when their spellings or character strings overlap heavily.
- (C) Words that appear in similar contexts tend to have similar meanings, so they receive similar vectors.
- (D) Each word must be hand-labeled with its correct sense before any embedding can be learned.

The distributional hypothesis ties meaning to context: words in similar contexts get similar vectors, which is exactly what word2vec exploits.

Problem 9

Which statement correctly characterizes multinomial Naive Bayes as a text classifier?

- (A) Naive Bayes is discriminative and learns $P(\text{class} \mid \text{text})$ directly through gradient descent over epochs.
- (B) Naive Bayes abandons conditional independence and explicitly models dependencies among all words.
- (C) Naive Bayes estimates $P(\text{class} \mid \text{text})$ first and applies Bayes' rule to recover the word likelihoods.
- (D) Naive Bayes is generative, assumes conditional independence given the class, and is fit by counting in one pass.

Naive Bayes is generative with a conditional-independence assumption and a single counting pass; it models $P(\text{text} \mid \text{class})$ and the priors, then applies Bayes' rule to obtain $P(\text{class} \mid \text{text})$.

Problem 10

A text classifier applies a sigmoid output unit to bag-of-words or averaged-embedding features. Which statement about this head is correct?

- (A) The sigmoid head is generative and models $P(\text{text} \mid \text{class})$ over the input features.
- (B) The sigmoid head over these features is equivalent to logistic regression on those features.
- (C) The sigmoid head needs the conditional-independence assumption in order to be trained.
- (D) The sigmoid head can only be fit by a single closed-form counting pass, like Naive Bayes.

A sigmoid applied to a weighted sum of features is logistic regression; it is discriminative, makes no independence assumption, and is fit by gradient descent.

Problem 11

Which statement correctly contrasts static and contextual word embeddings?

- (A) Contextual embeddings give “mouse” one fixed vector that is shared across every sentence.
- (B) Static embeddings give “mouse” a different vector in each sentence to resolve its ambiguity.
- (C) Static and contextual embeddings are identical because both are produced by Transformers.
- (D) A static embedding gives “mouse” one fixed vector, so the rodent and device senses collapse together.

Static embeddings (word2vec, GloVe) assign one vector per word, so the two senses of “mouse” merge; contextual embeddings from the Transformer family give a different vector per context.

Problem 12

A text classifier maps a vocabulary of 10,000 words to 32-dimensional vectors through an embedding lookup table, averages the embeddings of a sentence into one 32-dimensional vector (mean pooling), passes it through a fully connected hidden layer of 16 units (with bias and a ReLU activation), and ends in a single output unit (with bias). How many **trainable parameters** does the model have in total?

- (A) 320,000
- (B) 320,545
- (C) 320,528
- (D) 321,045

Step 1 (embedding table): $10,000 \times 32 = 320,000$ weights; mean pooling adds none.

Step 2 (hidden layer): 32×16 weights + 16 biases = $512 + 16 = 528$.

Step 3 (output unit): 16×1 weights + 1 bias = 17.

Step 4 (total): $320,000 + 528 + 17 = 320,545$.

Problem 13

Select All That Apply:

Which of the following statements about segmentation architectures are correct?

- (A) A fully convolutional network keeps dense layers at the end in order to output its per-pixel class map.
- (B) Semantic segmentation assigns a class to every pixel but does not distinguish two objects of the same class.
- (C) Instance segmentation produces a separate mask for each object, even when the objects share a class.
- (D) Mask R-CNN is a single-stage detector that produces no masks.
- (E) U-Net uses skip connections to carry fine spatial detail from encoder layers into the decoder's upsampling path.

Answer: (B), (C), (E). A fully convolutional network replaces the dense head with convolutions (A), and Mask R-CNN is a two-stage detector that does output per-object masks (D).

Problem 14

Select All That Apply:

Which of the following statements about Grad-CAM and attribution methods are correct?

- (A) Grad-CAM weights the last convolutional feature maps by their pooled class-score gradients, sums them, and applies a ReLU.

- (B) Because it is computed from gradients, Grad-CAM certifies that the model’s reasoning is correct.
- (C) A classifier that separates wolves from huskies using background snow is a desirable shortcut to encourage.
- (D) Grad-CAM heatmaps are coarse because their resolution matches the last convolutional layer, not the input image.
- (E) Occlusion-based attribution is model-agnostic: it needs only forward passes, not gradients.

Answer: (A), (D), (E). No attribution method certifies correctness (B), and the snow shortcut is an undesirable “Clever Hans” failure that Grad-CAM helps expose (C).

Problem 15

Select All That Apply:

Which of the following statements about text standardization and tokenization are correct?

- (A) A token must always be a whole word; subword and character units are not valid tokens.
- (B) Lowercasing and stripping punctuation cut near-duplicate tokens but can erase signal a task needs.
- (C) Stemming maps inflected forms such as “staring” and “stared” to a shared root.
- (D) A bag-of-bigrams captures some local word order that a bag-of-unigrams discards entirely.
- (E) Standardization should be applied identically for every task, regardless of what the task needs.

Answer: (B), (C), (D). Tokens may be words, subwords, or characters (A), and standardization is a task-dependent choice, not a fixed default (E): a question detector, for instance, should keep “?”.

Problem 16

Select All That Apply:

Which of the following statements about bag-of-words representations are correct?

- (A) A unigram bag-of-words maps the sentences “dog bites man” and “man bites dog” to the identical vector.

- (B) The length of a bag-of-words vector is set by the document length and is independent of the vocabulary.
- (C) Binary bag-of-words records only whether a term is present, ignoring how many times it occurs.
- (D) Bag-of-words vectors are typically dense, with most of their entries nonzero.
- (E) TF-IDF lowers the weight of terms that occur in almost every document of the corpus.

Answer: (A), (C), (E). A bag-of-words vector has one entry per vocabulary term, so its length is the vocabulary size, not the document length (B), and these vectors are sparse since any one document uses few of the words (D).

Problem 17

Select All That Apply:

Which of the following statements about word embeddings and word2vec are correct?

- (A) word2vec and GloVe capture a gender direction, so $\text{king} - \text{man} + \text{woman} \approx \text{queen}$.
- (B) Pretrained word2vec or GloVe vectors can be loaded when task data is scarce, a text analogue of transfer learning.
- (C) Two words with a high cosine similarity must also have nearly equal vector magnitudes.
- (D) Embeddings learned from human text reproduce social biases such as stereotyped gender-occupation associations.
- (E) Skip-gram learns its vectors self-supervised, with no hand-labeled (target, context) pairs required.

Answer: (A), (B), (D), (E). Cosine similarity measures direction, not magnitude, so a high cosine says nothing about the vectors' lengths (C).

Problem 18

Select All That Apply:

Which of the following statements about classifier evaluation metrics are correct?

- (A) The F_1 score is the arithmetic mean of precision and recall.
- (B) On an imbalanced corpus, accuracy can stay high even when recall on the minority class is near zero.

- (C) Precision is the fraction of predicted positives that are actually positive.
- (D) Recall is the fraction of actual positives that the model correctly retrieves.
- (E) Raising the decision threshold generally increases recall at the expense of precision.

Answer: (B), (C), (D). F_1 is the harmonic (not arithmetic) mean of precision and recall (A), and raising the threshold makes the model more conservative, which raises precision and lowers recall (E).

Problem 19

Fill in the blanks:

A detector is evaluated on an image containing exactly 3 ground-truth objects of one class. It returns 5 predicted boxes; ranked from highest to lowest confidence, each is either correct (a true positive) or incorrect:

Rank	1	2	3	4	5
Outcome	correct	correct	incorrect	correct	incorrect

For each prefix of the ranked list, precision is the number of correct boxes so far divided by the number of boxes so far, and recall is the number of correct boxes so far divided by 3. Average Precision is the mean of the precision values measured at the ranks where a new correct box (and hence a new recall level) appears.

- (a) The recall after rank 4 is **1** (i.e. 3/3). (b) The Average Precision for this image, to two decimal places, is **0.92**.

Step 1 (find the correct ranks): new correct boxes appear at ranks 1, 2, and 4.

Step 2 (precision at each): rank 1: $1/1 = 1$; rank 2: $2/2 = 1$; rank 4: $3/4 = 0.75$.

Step 3 (recall after rank 4): 3 of the 3 objects are now found, so recall = $3/3 = 1$.

Step 4 (average precision): $AP = \frac{1}{3}(1 + 1 + 0.75) = \frac{2.75}{3} \approx 0.92$.

Problem 20

Fill in the blanks:

In skip-gram with negative sampling, the loss contributed by one target word w together with one true context word c and one randomly drawn negative sample n is

$$L = -\left[\ln \sigma(\mathbf{c} \cdot \mathbf{w}) + \ln \sigma(-\mathbf{n} \cdot \mathbf{w})\right], \quad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

Suppose the learned vectors give $\mathbf{c} \cdot \mathbf{w} = \ln 3$ and $\mathbf{n} \cdot \mathbf{w} = \ln 3$.

- (a) $\sigma(\mathbf{c} \cdot \mathbf{w}) = \mathbf{0.75}$. (b) $\sigma(-\mathbf{n} \cdot \mathbf{w}) = \mathbf{0.25}$. (c) the loss L , to two decimal places, is **1.67**.

Step 1 (positive term): $\sigma(\ln 3) = \frac{1}{1 + e^{-\ln 3}} = \frac{1}{1 + 1/3} = \frac{3}{4} = 0.75$.

Step 2 (negative term): $\sigma(-\ln 3) = \frac{1}{1 + e^{\ln 3}} = \frac{1}{1 + 3} = \frac{1}{4} = 0.25$.

Step 3 (loss): $L = -[\ln 0.75 + \ln 0.25] = -\ln(0.75 \times 0.25) = -\ln(0.1875) \approx 1.67$.